

A Unified Approach to Back-EMF Detection for Brushless DC Motor Drives without Current and Hall Sensors

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Abstract— The main theme of this paper is to present a unified approach to back-EMF detection of brushless DC motor drives without using any current and Hall sensors. Theoretical analysis of back-EMF detection is presented, and followed by the relationship between PWM techniques and back-EMF detection. It will be shown that the back-EMF detection depends upon the PWM techniques and the method is required to be slightly modified as the PWM technique is changed. Experimental results derived from BLDCM drives *without* using any current and Hall sensors fully confirm the theoretical analysis.

Keywords—pulse-width modulation; brushless DC motor drives

I. Introduction

As compared to DC motor, Brushless DC Motor (BLDCM) is with the advantages of higher power density, and no need of mechanical commutation mechanism, which results in compact and robust structure. In comparison with induction motor, BLDCM doesn't have copper losses on the rotor side. Because of these features, BLDCM becomes more popular for the applications where efficiency is a critical issue or spike caused by mechanical commutation is not allowed. However, some technical issues for BLDCM control have drawn our attention. The topics include how to generate commutation signals without using expensive sensor for the realization of BLDCM drives.

For BLDCM drives control, zero-crossing point detection of back Electron-Magnetic Force (EMF) becomes one of the most essential issues. Several approaches to zero-crossing detection of back-EMF to achieve commutation control have been proposed. These methods include using position sensors, e.g. Hall sensors [1], or current sensor [2], center-tap voltage and a resistor in series with DC-link voltage source [3]. In [4], terminal voltages of BLDCM and filter are used for zero-crossing detection. The filter used to remove the high switching frequency noise signals caused by PWM control results in time delay and thereby deteriorating the performance of torque response. In [5], the zero-crossing of back-EMF is detected by the *first* non-zero current pulse after the free-wheeling period of the floating phase. Several independent voltage sources for detection circuit and proper isolation mechanisms are required in this approach. Similar principle is applied to derive the zero-crossing point of back-EMF as shown in [6-9].

However, these approaches are not suitable for very *high* speed operation. This problem has been solved by the approach shown in [10], however, the approach is not suitable for *low* speed or low duty ratio applications. To the

authors' best knowledge, the relationship between PWM techniques and back-EMF detection method has not yet been fully investigated.

In this paper, a unified back-EMF detection approach for various kinds of PWM techniques will be presented. Theoretical analysis of back-EMF detection is presented, and followed by the relationship between PWM techniques and back-EMF detection. It will be shown that the back-EMF detection depends upon the PWM techniques. And the back-EMF detection methods for both low and high duty ratio applications are also investigated. Experimental results derived from experimental BLDCM drives *without* using any current and Hall sensors fully confirm the theoretical analysis.

II. PWM Techniques

The performance of the BLDCM drives is decided by the commutation control techniques. Fig. 1 illustrates the PWM techniques for commutation control of BLDCM drives using one of the inverter legs as an example.

As shown in Fig. 1 (A), the high-side power device is controlled by chopper signal every consecutive 120 degrees in a fundamental period and the associated low-side control signal is shifted by 180 degrees, as compared to its high-side one, to clamp the related inverter output to the negative DC-link rail. The control signals for the other two legs are shifted by 120 degrees and 240 degrees, respectively.

For PWM technique 2, it turns high-side power device on and lasts for one-sixth fundamental period. In the following 60 degrees, the high-side power device is controlled by chopper signal. The same control signal is applied to the associated low-side power device except 180-degree phase shift.

For PWM technique 3, the high-side power device is chopped in one-sixth fundamental period. In the following 60 degrees, the high-side power device is turn on and clamped to the positive DC-Link rail.

As shown in Fig. 1 (D) for PWM technique 4, the chop-controlled area for high-side power device is divided into two parts, each lasts for 30 degrees. The same control signal is applied to the associated low-side power device except 180-degree phase shift.

Fig. 1 (E) shows the PWM control signals for PWM technique 5. Basically, this PWM technique is the same as PWM technique 1. However, only during free-wheeling period, the high side power device is clamped to positive DC-Link rail, and the low side power device is with PWM control.

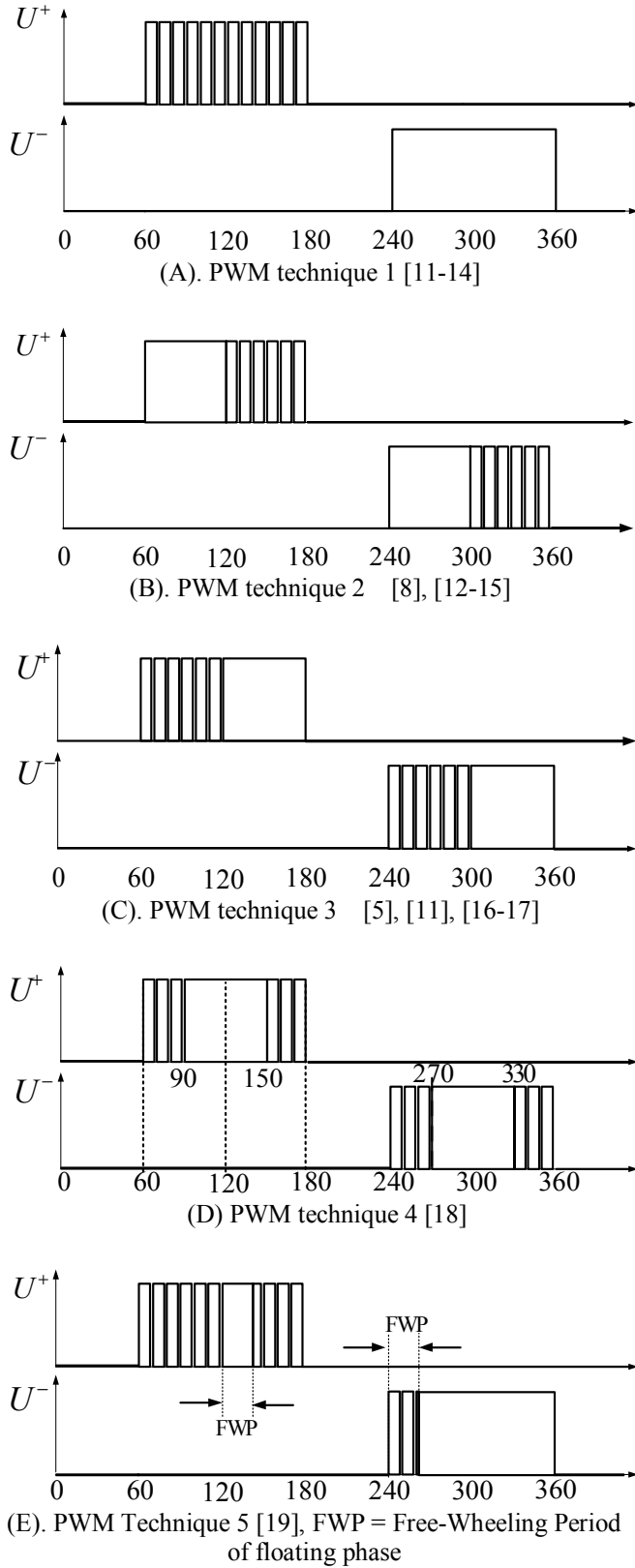


Fig. 1 Summary of commutation control techniques for BLDCM drives

III. Unified Back-EMF Detection Approach

Fig. 2 defines the terminal voltages of three-phase windings. As shown in Fig. 2, “ v_p ” denotes the terminal voltage for the phase connected to the positive DC-link rail during PWM control period. “ v_n ” indicates the terminal voltage for the phase connected to the negative DC-link rail during PWM control period. “ v_o ” is the terminal voltage for the floating phase. “PWM control” means the chopper signal is “ON”. This chopper on period is decided by the duty. The back-EMF voltage is detected via the terminal voltage of floating phase.

Since the back-EMF is detected from the terminal voltage of floating phase, the following equations can be derived:

$$i_p = -i_n \text{ and } i_o = 0 \quad (1)$$

By Fig. 2 and Kirchhoff's Voltage Law,

$$\begin{aligned} v_n &= v_p - (i_p r_s + L_s \frac{di_p}{dt}) - e_p \\ &= v_n + (i_p r_s + L_s \frac{di_p}{dt}) - e_n \end{aligned} \quad (2)$$

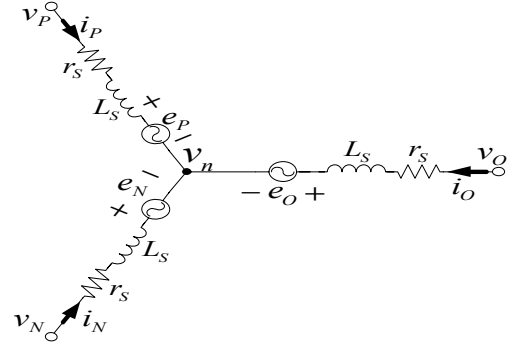


Fig. 2 Definition of terminal voltages of BLDCM

By (2), the central-tap voltage is derived as:

$$v_n = \frac{v_p + v_n}{2} - \frac{e_p + e_n}{2} \quad (3)$$

By Fig. 2, the terminal voltage of the floating phase is shown as:

$$\begin{aligned} v_o &= v_n + e_o \\ &= \frac{v_p + v_n}{2} - \frac{e_p + e_n}{2} + e_o \end{aligned} \quad (4)$$

Since

$$e_p + e_n = 0 \quad (5)$$

Equation (4) can be rewritten as:

$$v_o = \frac{v_p + v_n}{2} + e_o \quad (6)$$

As shown in (6), the terminal voltage of the floating phase depends upon the voltage applied to the phase connected to the positive and negative DC-link rails during PWM control period.

Fig. 2 illustrates the PWM control signals and terminal voltage using PWM technique 2 as an example. As shown in Fig. 2, the terminal voltage of the floating phase has either rising edge or falling edge. For both, when the terminal voltage becomes zero, the zero-crossing point occurs.

For the case of rising edge shown in Fig. 3 (A), the terminal voltage of floating phase is analyzed when chopper signal is “ON” and “OFF”. In the “Chopper ON” period for PWM technique 2, “positive phase” is connected to positive DC-link rail and “negative” phase is connected to negative DC-link rail. The terminal voltage for the floating phase is derived by (6) and can be written as:

$$v_o = \frac{V_{DC}}{2} + e_o \quad (7)$$

Under this circumstance, the condition for zero-crossing point detection is:

$$v_o = \frac{V_{DC}}{2} \quad (8)$$

In the “Chopper OFF” period for PWM technique 2, “positive phase” and “negative phase” are connected to negative DC-link rail. The terminal voltage for the floating phase is derived by (6) and can be written as:

$$v_o = e_o \quad (9)$$

Fig. 3 (B) shows the terminal voltage of flowing phase for the case of falling edge. In the “Chopper ON” period for PWM technique 2, “positive phase” is connected to positive DC-link rail and “negative” phase is connected to negative DC-link rail. The terminal voltage for the floating phase is derived by (6) and can be written as:

$$v_o = \frac{V_{DC}}{2} + e_o \quad (10)$$

Under this circumstance, the condition for zero-crossing point detection is:

$$v_o = \frac{V_{DC}}{2} \quad (11)$$

As shown in Fig. 3 (B), in the “Chopper OFF” period for PWM technique 2, “positive phase” and “negative phase” are connected to positive DC-link rail. The terminal voltage for the floating phase is derived by (6) and can be written as:

$$v_o = V_{DC} + e_o \quad (12)$$

And therefore, the terminal voltage of the floating phase is “ V_{DC} ” when zero crossing occurs. Therefore, we can summarize that there are three kinds of voltage combinations for the zero-crossing point detection. As shown in (6), the terminal voltage of the floating phase

depends upon the voltage applied to the phase connected to the positive and negative DC-link rails during PWM control period. Table 1 summarizes the back-EMF detection conditions for these three cases. As shown in Table, the conditions for zero-crossing point detection are decided by the PWM control techniques and PWM control status, which is with “Chopper ON” or “Chopper OFF”. The PWM control techniques decide the voltage level of the non-floating phases.

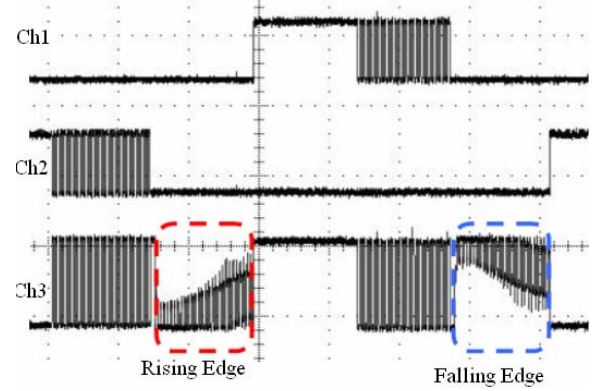
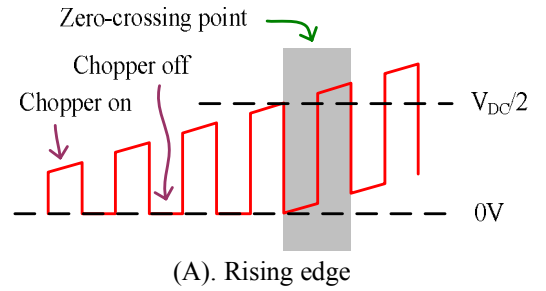
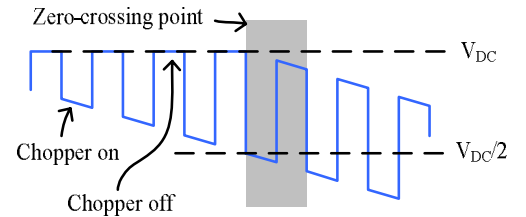


Fig. 2 Measured results, PWM control signals and terminal voltage, Ch 1 = High-side signal, Ch 2 = Low-side signal, Ch 3 = Terminal voltage



(A). Rising edge



(B). Falling edge

Fig. 3 Terminal voltage

Table 1 Zero-Crossing Point (ZCP) detection of back EMF

Case	1	2	3
Positive side v_P	V_{DC}	V_{DC}	0V
Negative side v_N	V_{DC}	0V	0V
Open phase v_O	$V_{DC} + e_o$	$\frac{V_{DC}}{2} + e_o$	e_o
Conditions for ZCP Detection	$v_o = V_{DC}$	$v_o = V_{DC} / 2$	$v_o = 0$
Occurs	Chopper Off	Chopper On	Chopper Off

It is worthy of noting that the allowable time interval for back-EMF detection using “0 V” as reference voltage, which equals to $(1-D)T_s$, D = duty and T_s = sampling period, is significantly reduced for very high duty ratio. In contrast, when $0.5 V_{dc}$ is used as the reference for back-EMF detection, the back-EMF detection can work well only for a certain duty-ratio range. However, this approach can't work at very low duty-ratio range. To work properly in both high and low duty ratio region, zero-crossing points of back-EMF are considered at “Chopper on” and “Chopper off”, respectively.

Similar results for the conditions of zero-crossing detection and detection instant, either “Chopper On” or “Chopper Off”, of the other PWM techniques can be derived. Table 2 summarizes the back-EMF detection method for the associated PWM techniques. As shown in Table 2, the back-EMF detection method indeed depends upon the PWM techniques. Moreover, to achieve full duty range detection, the terminal voltage should be sensed either chopper control signal is “ON” or “Off”. The detection instance is also decided by the PWM techniques.

Table 2 Back-EMF detection method for PWM techniques

PWM	Back-EMF detection algorithm
PWM 1	Rising & Falling edges: $v_o = V_{DC}/2$, Chopper on, duty $\geq 50\%$ Rising & Falling edges: $v_o = 0$, Chopper off, duty $< 50\%$
PWM 2	Rising & Falling edges: $v_o = V_{DC}/2$, Chopper on, duty $\geq 50\%$ Rising edge: $v_o = 0$, Chopper off, duty $< 50\%$ Falling edge: $v_o = V_{DC}$, Chopper off, duty $< 50\%$
PWM 3	Rising & Falling edges: $v_o = V_{DC}/2$, Chopper on, duty $\geq 50\%$ Rising edge: $v_o = V_{DC}$, Chopper off, duty $< 50\%$ Falling edge: $v_o = 0$, Chopper off, duty $< 50\%$
PWM 4	Rising & Falling edges: $v_o = V_{DC}/2$, Chopper on
PWM 5	Rising & Falling edges: $v_o = V_{DC}/2$, Chopper on, duty $\geq 50\%$ Rising & Falling edges: $v_o = 0$, Chopper off, duty $< 50\%$

IV. Experimental Confirmation

The experimental system is set up using FPGA as the controller to realize the sensorless control technique. The control technique includes zero-crossing point of back-EMF detection, PWM control and duty ratio control. Table 3 shows the features of the FPGA used in deriving the experimental results.

Fig. 4 shows the block diagram of the experimental system, which consists of an FPGA controller, inverter, BLDCM (the specifications are included in the Appendix), and back-EMF detection circuit. To confirm the back-EMF detection algorithm, neither current nor Hall sensor is required.

Table 3 Features of FPGA

Part number	EPF10K70RC240-3
Typical gates	70000
Logic Elements	3744
I/O Pin Count	189
Supply Voltage	5V

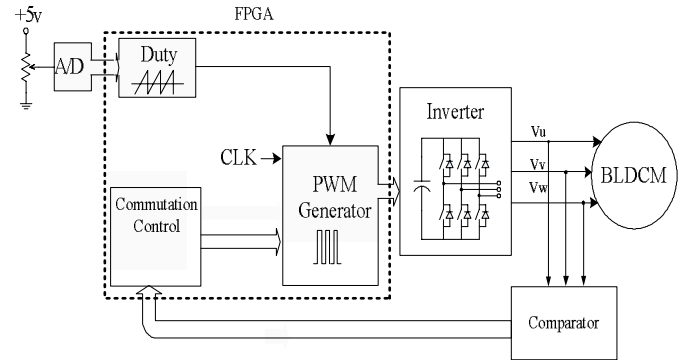
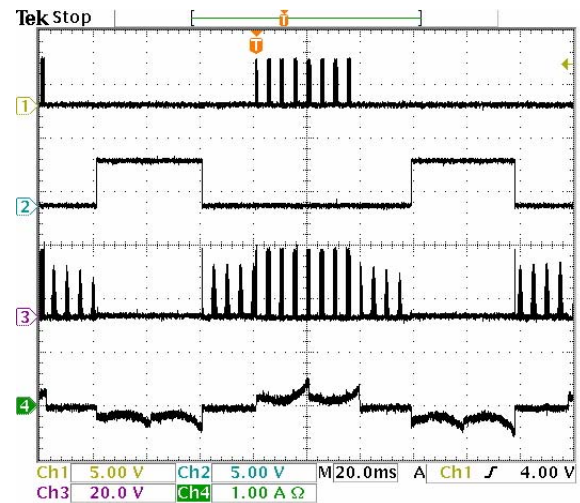
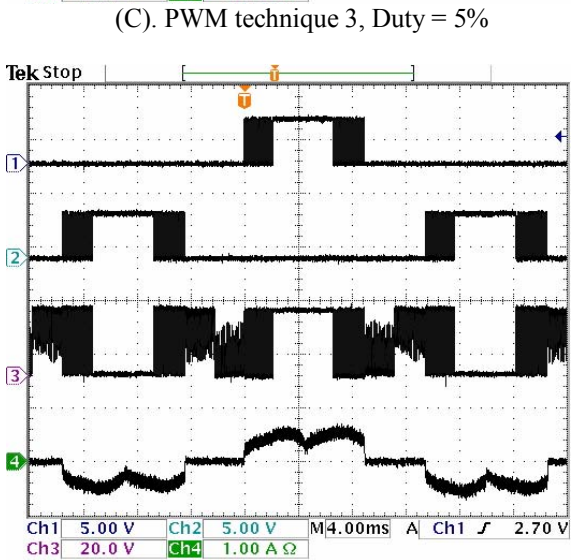
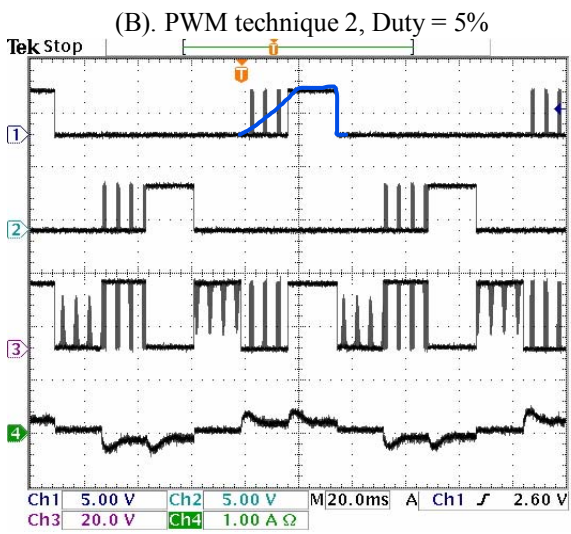
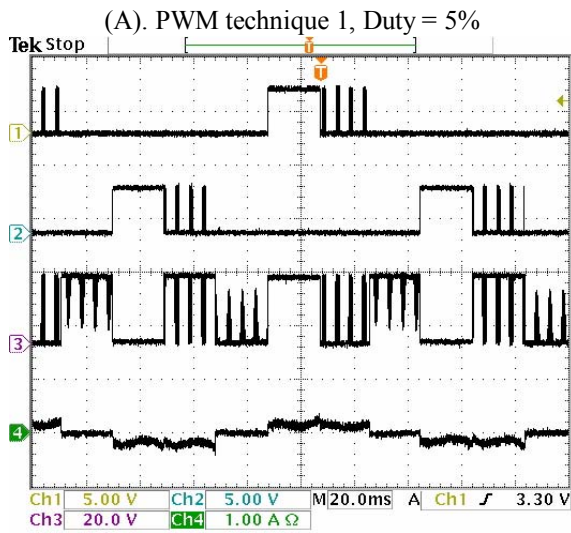


Fig. 4 Block diagram of the sensorless experimental system

Fig. 5 and Fig. 6 show the experimental results of PWM waveforms for hi-side and low-side power devices, voltage waveform and current waveform when at low and high duty ratios, respectively. As confirmed in Fig. 5, for PWM 4, the back-EMF can only be detected under the condition of “Chopper On”, the operation of low duty is therefore limited. In contrast, for the other PWM techniques, the motor can operated for duty ratio = 5% because back EMF can be detected under the condition of “Chopper Off”.

As shown in Fig. 5 and Fig. 6, the waveforms of back-EMF change as the PWM techniques are modified. Moreover, the zero-crossing point of back-EMF can be detected using the algorithm at both high and low duty ranges. These results confirm the presented algorithm.





(D). PWM technique 4, Duty = 20%

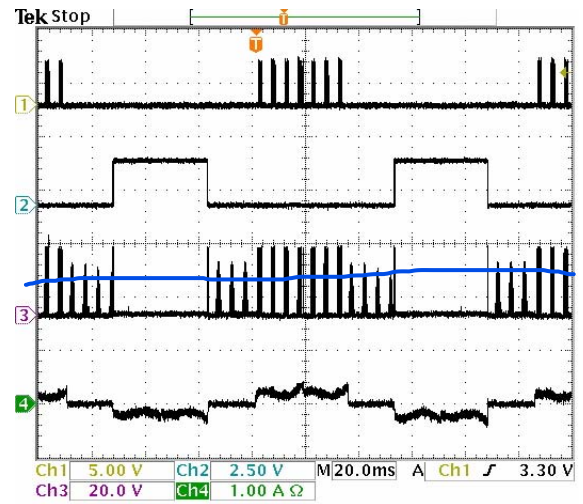
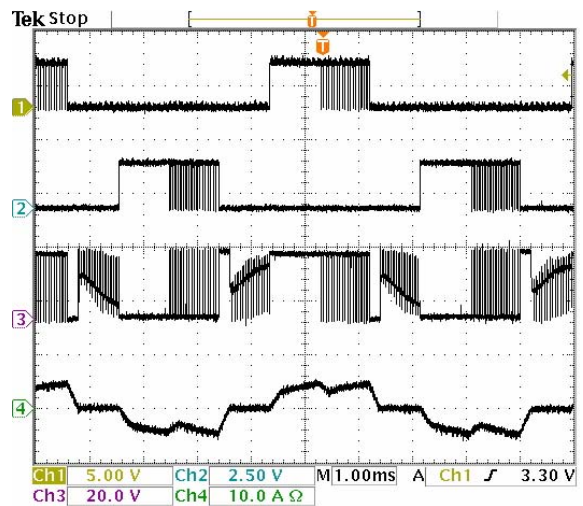
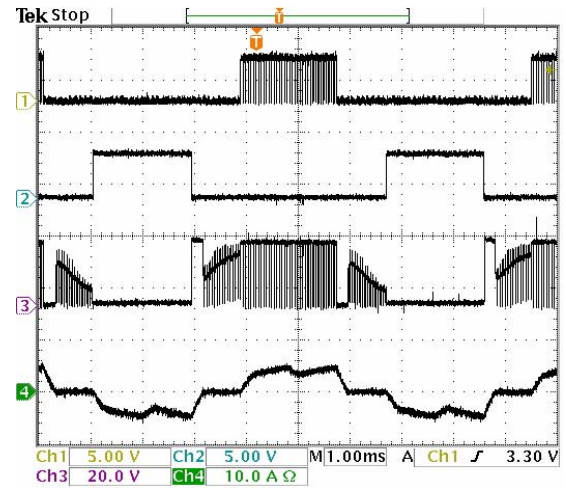
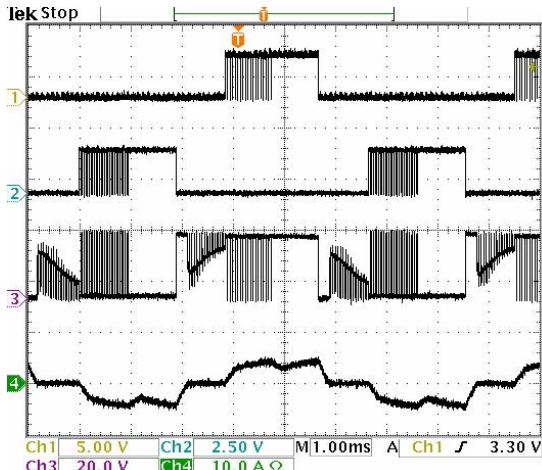
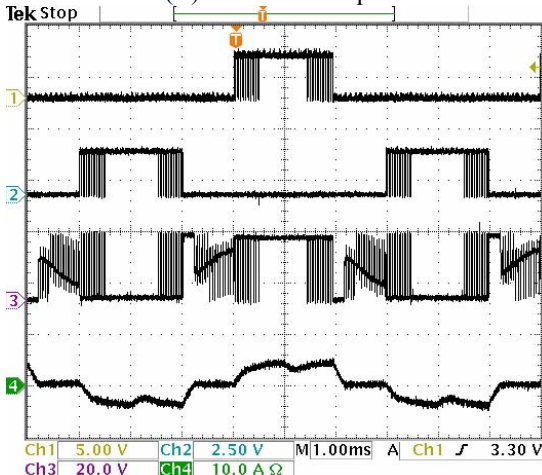


Fig. 5 Experimental results, CH1: High-side control, CH2: Low-side control, CH3: phase voltage, CH4: phase current

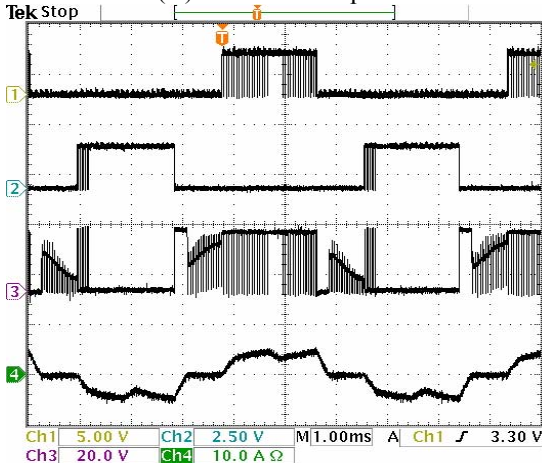




(C). PWM technique 3



(D). PWM technique 4



(E). PWM technique 5

Fig. 6 Experimental results, CH1: High-side control, CH2: Low-side control, CH3: phase voltage, CH4: phase current, duty = 90 %

V. Conclusions

The contribution of this paper is to present a unified back-EMF detection method for PWM-controlled brushless DC motor drives. It is shown by theoretical analysis that

the back-EMF detection method is required to be slightly modified as the PWM control method is changed. Experimental results derived from BLDCM drives *without* using any current and Hall sensors fully confirm the theoretical analysis.

Appendix: Motor specifications

No. of poles = 8, rated power = 70 W, Vdc = 24 V, Rated speed = 2500 rpm

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